HASEEB MIR

Hardware Systems Engineering | Power Electronics | Embedded Control & Validation

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CORE TECHNICAL EXPERTISE

Power Electronics & System Architecture: High-power AC–DC and DC–DC conversion, SiC/GaN power converters, Totem-Pole PFC, LLC resonant converters, DAB (TPS modulation), PSFB, flyback converters, solar microinverters, power-stage optimization, and topology evaluation.

Power Hardware & Magnetic Design: Schematic capture, multilayer PCB design and layout, gate-drive circuits, analog/digital protection, voltage and current sensing, high-frequency magnetic design, transformer and inductor optimization, and auxiliary power supplies.

Control Systems & Embedded Development: Digital closed-loop control, small-signal modeling, frequency-response analysis, compensator design, stability assessment, model-predictive control (MPC), MPPT, embedded C, TI C2000 DSPs, real-time control firmware, and CAN communication.

System Validation & Characterization: Hardware bring-up, power-stage validation, double-pulse testing, SiC/GaN switching-loss characterization, thermal analysis, transient-response testing, efficiency measurements, sensor calibration, fault troubleshooting, and laboratory instrumentation.MATLAB/Simulink, LTspice, QSPICE, PSIM, ANSYS Maxwell, FEMM, KiCad, Altium Designer, Python, and numerical optimization.

FPGA & Industrial Communication: Xilinx Zynq-7020, FPGA-based acquisition, embedded Linux, Ethernet, IEC 61850 Sampled Values, IEEE 1588 PTP, DMA, and hardware synchronization.

PROFESSIONAL EXPERIENCE

ChargeWorks (Incubated at IISc), Bengaluru
Feb 2026 – Present

Senior System Engineer

Project 1: 3.3 kW SiC-Based AC–DC On-Board Charger

- Led end-to-end development of a **3.3 kW SiC-based AC–DC charger** incorporating bridgeless Totem-Pole PFC and LLC resonant DC–DC conversion stages.
- Designed, implemented, and experimentally validated **digital closed-loop control**, successfully demonstrating operation **up to 3.3 kW**.
- Performed **frequency-response analysis, compensator design, loop-stability assessment, and transient-response optimization** using MATLAB/Simulink and TI C2000 DSPs.
- Developed power-stage schematics, PCB layouts, sensing interfaces, gate-drive circuitry, and hardware protection systems for high-frequency SiC-based conversion.
- Performed **switching-loss characterization, thermal analysis, power-stage bring-up, and experimental hardware validation**.

Project 2: 2.2 kW GaN-Based AC–DC Charger

- Developed a **2.2 kW GaN-based AC–DC charger** incorporating Totem-Pole PFC and an optimized LLC resonant converter with **EPC GaN synchronous rectification** on the secondary side.
- Evaluated **Dual Active Bridge (DAB)** and LLC converter architectures. Developed **Triple Phase Shift (TPS) modulation** and formulated an **RMS-current minimization problem** to reduce conduction losses and current stress.
- Conducted comparative topology analysis based on efficiency, current stress, operating characteristics, and implementation tradeoffs; selected the **LLC architecture** for the charger.
- Optimized the **LLC resonant tank**, including resonant inductance, magnetizing inductance, capacitance, transformer design, switching-frequency range, and device stress.
- Developed schematic designs, PCB layouts, magnetic components, GaN gate-drive circuits, and digital control strategies supported by simulation and hardware testing.
- Achieved best-case measured efficiencies of approximately **97.3% for the BTPFC stage** and **98.8% for the LLC stage**.

Project 3: Digital Control Hardware and Auxiliary Power Systems

- Designed and validated a custom **TI TMS320F28379D control card**, including schematic capture, multilayer PCB layout, ADC signal conditioning, sensor interfaces, protection circuitry, and gate-drive control interfaces.
- Developed modular **embedded C firmware** for PWM generation, ADC acquisition, sensor calibration, closed-loop regulation, converter state machines, fault handling, and **CAN communication**.
- Designed and experimentally validated high-frequency **inductors and transformers** for power-converter prototypes.
- Developed a **25 W multi-winding flyback auxiliary power supply** with RCD clamping, transformer design, and analog-IC-based closed-loop regulation.

Indian Institute of Science (IISc), Bengaluru
Mar 2023 – Feb 2026

Senior Research Associate – Sustainable Power Electronics Lab

Project 1: FPGA based Merging Unit for Digitsl Substation– PGCIL Funded

- Led hardware and embedded systems development of a **Digital Substation Merging Unit**, funded by **Power Grid Corporation of India Limited (PGCIL)** for India's first digital substation project at **Navsari, Gujarat**.
- Designed and validated **16-bit ADC-based voltage and current sensing hardware**, including analog signal conditioning,

schematic capture, PCB development, and precision data-acquisition interfaces.

- Developed a deterministic acquisition and processing architecture using **Xilinx Zynq-7020 FPGA**, integrating programmable logic, processor interfaces, and DMA-based data transfer.
- Implemented **IEEE 1588 PTP** synchronization using embedded Linux, hardware timestamping, and FPGA-based timing architecture.
- Developed **IEC 61850-9-2 Sampled Values** streaming at 4 kHz using lwIP Ethernet, real-time packetization, and communication validation.
- Implemented **AES-128 FPGA hardware encryption** and performed system-level validation using Wireshark, timing measurements, and hardware debugging.

Project 2: Model Predictive Control of Power Converters

- Developed and experimentally validated **Model Predictive Control (MPC)** for a 500W buck converter to improve transient response and dynamic regulation.
- Designed and implemented **MPC for Boost PFC converters**, including mathematical modeling, real-time embedded implementation, and hardware validation on **TI C2000 DSPs**.
- Performed MATLAB/Simulink modeling, controller evaluation, laboratory testing, and simulation-to-experiment verification.

Project 3: Reluctance Motor Electromagnetic Modeling and Nonlinear Control

- Developed a **Finite Element Method (FEM)-based electromagnetic model** of a reluctance motor using **FEMM**, capturing nonlinear magnetic characteristics and motor operating behavior.
- Developed a **nonlinear dynamic motor model** using FEMM-derived magnetic data and implemented a **closed-loop control strategy** with a unidirectional H-bridge power-converter topology.
- Performed simulation-based validation against published experimental results from a Transactions journal paper, achieving approximately **3–5% deviation** from reported hardware measurements.
- Analyzed electromagnetic and dynamic performance through coupled magnetic modeling, power-converter simulation, and nonlinear closed-loop control evaluation.

KEC International Ltd., India Jul 2022 – Feb 2023
Assistant Manager – Electrical Projects and Cables

- Supported engineering, testing, and commissioning activities for the **765/400/220 kV digital substation** at Navsari, Gujarat, including technical coordination and project execution.
- Gained cross-functional experience in **project costing, technical estimation, and techno-commercial evaluation**, working with engineering, sales, and project execution teams.
- Collaborated with **sales teams, customers, and equipment manufacturers** to communicate technical specifications, clarify engineering requirements, and coordinate project-related technical discussions.
- Supported **customer technical inquiries, LT/HT cable-system engineering, and vendor coordination**, including technical documentation and compliance with IEC/IS standards.

Arka Energy Pvt. Ltd., Bengaluru Jan 2022 – Jul 2022
R&D Engineer – Power Electronics

Project: 1.6 kW Solar Microinverter Development

- Developed a **1.6 kW solar microinverter system**, integrating MPPT-controlled DC–DC conversion and grid-connected DC–AC conversion.
- Designed and validated flyback converter hardware, high-frequency transformers, magnetic components, and power-stage circuitry.
- Developed **MPPT algorithms, grid synchronization, current control, and space-vector modulation** for DC–AC inverter operation.
- Performed MATLAB/Simulink modeling, controller design, digital firmware implementation using **TI F28379D**, and hardware validation.

J&K Government Contractors, Jammu & Kashmir Jul 2017 – Aug 2019
Junior Engineer

- Supervised the execution of **electrical installation works for a government dispensary**, coordinating on-site activities, contractors, and installation requirements.
- Supported the execution of **government-funded electrical infrastructure projects**, including installation of electrical distribution poles and overhead distribution lines.
- Performed **site supervision, material estimation, quantity verification, and project documentation**, coordinating with contractors and field teams to ensure timely project execution.
- Monitored installation progress, workmanship, and adherence to project specifications and electrical safety requirements.

EDUCATION

Master of Technology (M.Tech) 2021 – 2023
Power Electronics & Electric Drives
National Institute of Technology (NIT), Srinagar

Bachelor of Technology (B.Tech) 2012 – 2016
Electrical Engineering
Islamic University of Science and Technology (IUST), Awantipora, J&K

Diploma in Information Technology 2016 – 2017
Intech Technologies, Srinagar, J&K